

Amir R. Isazadeh

R&D Mechanical Engineer • Automated DFM Pipelines • Parametric CAD Design • CAE/FEA Simulation

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Professional Summary

R&D Mechanical Engineer specializing in **parametric CAD** (SolidWorks), **procedural 3D modeling** (Python), and **CAE/FEA** to drive product development from initial concept to manufacturable reality.

Engineered automated CAD/CAE architectures that collapse **product iteration cycles from days to hours**; a rapid and robust solution for complex design and optimization.

Expert in **bridging predictive simulation with DFM constraints**; a unique background in precision custom-fit device manufacturing and high-volume product/tooling optimization.

Proven track record of **mitigating risks** through rigorous prototyping, **validation**, and guiding cross-functional teams and external vendors to deliver **robust hardware/software solutions**.

Experience

R&D Mechanical and Biomedical Engineer

Edmonton, Alberta

Head and Neck Simulation Center, University of Alberta

2023–2025

- Engineered a **novel procedural CAD and CAE pipeline** for custom-fit surgical planning optimizations; an end-to-end virtual testbed to systematically resolve multi-variable mechanical constraints for **data-driven** surgical decision making; open-sourced as VSP-BioSim and published in PLOS ONE.
- Delivered **100× performance acceleration** by engineering a custom **FEM solver** (Python, Cython/C++) for highly non-linear materials which can **predict material failure modes** and mitigate operative risks; a validated framework adopted at two major hospitals in Alberta; open-sourced as hiFEM and published in PMB and JCARs.
- Automated complex geometry processing** and multi-component assembly generation by orchestrating 15+ modeling and FEA libraries. Achieved **99% pipeline reliability**, validated during a 20-hour continuous batch deployment.
- Validated sub-millimeter surgical precision** through physical benchtop test plans; leveraged SLA 3D-printing and toolkit prototyping, silicon die casting, and HD surface scanning to **guarantee manufacturability** and real-world compliance; published in SO.
- Achieved a **16× acceleration in mesh generation** by deploying an **AI-powered segmentation pipeline** (PyTorch, scikit-learn) to process raw 3D scan data into high-fidelity models; reduced lead times from 4 hours to 15 minutes to unlock same-day prototyping and deployment.
- Engineered **advanced mechanical design** of multi-component, manufacturable toolkits to translate virtual simulations into **high-precision hardware** for reliable execution of digital plans.
- Directed cross-disciplinary product development** alongside surgeons; distilled 40+ hours of field observations into actionable insights; **aligned stakeholders and external vendors** on critical design trade-offs via **iterative prototype validation** and continuous feedback loops.

Mechanical Engineer

Tehran

Center of Excellence in Design and Automation

2020–2021

- Engineered an **automatic DFM framework** (MATLAB, SolidWorks, FEA) to rapidly optimize complex tool and part designs for high-volume manufacturing (sheet metal stamping and die design); published in JEM.
- Developed a **custom inverse FEA solver** to predict material limitations and **mitigate failure risks** (tearing, wrinkling) for mass-production sheet metal stamping; validated a **90% reduction in defect zones** and bypassed costly physical trial-and-error; published in AMM.
- Executed rigorous test plans on physical prototypes and **validated** algorithmic predictions against metrology data (**≈ 3% error margin**) to ensure physical compliance and **reliable mass manufacturability**.
- Architected a cohesive **parametric CAD** (SolidWorks) to **FEA** (ABAQUS) pipeline capable of generating and

evaluating dozens of design variations; **10× Faster R&D Iteration Cycles**; streamlined the management of CAD assets and resolved production constraints prior to physical prototyping.

- **Directed a cross-disciplinary team** in deploying a parametric CAD and DFM workflow to IKCO R&D (automotive supplier); optimized tool assemblies and sheet designs for production lines with a projected $\approx$ **35% defect reduction**; managed trade-offs between delivery schedule, cost, and manufacturing limits.

Skills & Expertise

- **Mechanical Design & CAD:** Parametric CAD (SolidWorks), Assembly Design, Fusion 360, Inventor, Procedural 3D Modeling and Geometric Processing (Python, Blender, Rhino/Grasshopper, Point Clouds, Open3D).
- **Mass Production & DFM:** Sheet Metal Stamping, Deep Drawing, Injection Molding & Die Casting, High-Volume Tooling, and Rapid Prototyping.
- **Hardware Validation:** System Mockups, Physical Test Plan Specification, Performance Reliability Testing, 3D Surface Scanning/Metrology.
- **CAE, Simulation & Material Mechanics:** Non-Linear FEA (ABAQUS, PolyFEM), Material Failure Prediction, Contact Mechanics, Large Deformations, Soft Tissue Biomechanics.
- **Data Analysis & Visualization:** NumPy, SciPy, Matplotlib, Seaborn, ParaView, PyVista, trame-vtk, DoE, ANOVA, statsmodels, R, STATA.
- **Automation & Data:** Python, PyTorch, scikit-learn, C++, MATLAB, Git, GitHub, Multi-Objective Optimization (GA), SQL (PostgreSQL, SQLite), NoSQL (MongoDB, Firebase), REST APIs.

Education

PhD in Biomedical Engineering

University of Alberta | 2023–2025

- Patient-specific virtual surgical planning: Procedural Modeling & Simulation. **GPA: 3.95/4.0**
- Secured **\$270K** in competitive funding, mentored junior researchers, and completed PhD in under 3 years.

M.Sc. in Mechanical Engineering

Sharif University of Technology | 2019–2021

- Mass Production DFM: Rapid optimization of part manufacturing processes. **GPA: 4.0/4.0**
- Eligible for registration as a Professional Engineer (P.Eng).

B.Sc. in Mechanical Engineering

Amirkabir University of Technology | 2013–2018

- Parametric design of axial-flow compressors via procedural modeling. **GPA: 3.9/4.0**

Selected Publications

1. Guided tongue reconstruction: Feasibility analysis of additively manufactured surgical toolkit on a silicone tongue model (2026). *Surgical Oncology*. doi: 10.1016/j.suronc.2026.102384 (**1st author**).
2. Computational analysis of tongue reconstruction surgery: The impact of free flap site and volume on the tongue anatomy and biomechanics (2026). *PLOS ONE*. (under review) (**1st author**).
3. Patient-specific virtual surgical planning for tongue reconstruction: Evaluating hyperelastic inverse FEM with four simulated tongue cancer cases (2025). *Phys Med Biol*. doi: 10.1088/1361-6560/adebd8 (**1st author**).
4. Optimum design of middle-stage tool geometry and addendum surfaces in sheet metal stamping processes using a new isogeometric-based framework (2022). *JEM*. doi: 10.1177/09544054211041046 (**2nd author**).
5. Some improvements on the one-step inverse isogeometric analysis by proposing a multi-step inverse isogeometric methodology in sheet metal stamping processes (2021). *APM*. doi: 10.1016/j.apm.2020.09.032 (**1st author**).

Selected Recognition & Awards

- Graduate Student Support Award (\$10,000): University of Alberta 2024
- Nominated for Vanier Canada Graduate Scholarships: CIHR Federal Competition 2023
- Faculty of Rehabilitation Medicine GRAF (\$64,000): University of Alberta 2023
- Joint Biomechanics Scholarship (\$99,000): Queensland University of Technology (awarded, declined) 2022
- Research Tuition Fee Sponsorship (\$96,000): Queensland University of Technology (awarded, declined) 2022
- Recognized as “Outstanding Student” and ranked in the top 5% of graduates: Sharif University of Technology 2021
- Ranked 11th (out of 13,900) in the national M.Sc. entrance exam: NOET, Tehran 2018